WORKED SOLUTION

STEP III 1989, Question 9

Pure • Summing Power Series

STEP III 1989, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q9

The key idea

Each series is a familiar power series $\sum a_r x^r$ evaluated at a particular x, so the job is to spot which one. The geometric series $\frac{1}{1-x} = 1 + x + x^2 + \cdots$ is the parent of the first two. Differentiating it brings down a factor r, and integrating it divides by r. For the third, the products of odd numbers over factorials are what the binomial expansion with a negative fractional power produces.

Solution

Throughout we use, for $|x| < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{r=1}^{\infty} x^{r-1}.$$

The question tells us not to worry about convergence, so we freely differentiate and integrate term by term. (All three values of x we use lie well inside the interval of convergence, so this is in fact legitimate.)

Part (i)

The general term is $r\left(\frac{1}{2}\right)^{r-1}$, so we want $\sum r x^{r-1}$ at $x = \frac{1}{2}$. Differentiating the geometric series term by term,

$$\sum_{r=1}^{\infty} r x^{r-1} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2}.$$

Putting $x = \frac{1}{2}$:

$$1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \cdots = \frac{1}{\left(\frac{1}{2}\right)^2} = 4.$$

Part (ii)

The general term is $\frac{1}{r}\left(\frac{1}{2}\right)^{r-1}$. Dividing by r is undone by differentiation, so this time we integrate. Integrating the geometric series from 0 to x,

$$\sum_{r=1}^{\infty} \frac{x^r}{r} = \int_0^x \frac{\mathrm{d}t}{1-t} = -\ln(1-x).$$

The powers here are one too high, so we divide by x (for $x \neq 0$):

$$\sum_{r=1}^{\infty} \frac{1}{r} x^{r-1} = -\frac{\ln(1-x)}{x}.$$

Putting $x = \frac{1}{2}$:

$$1 + \frac{1}{2}\times\frac{1}{2} + \frac{1}{3}\times\frac{1}{2^2} + \cdots = -\frac{\ln\frac{1}{2}}{\frac{1}{2}} = 2\ln 2.$$

Part (iii)

Write

$$c_k = \frac{1\times 3\times\cdots\times(2k-1)}{k!}, \qquad c_0 = 1.$$

Ask yourself Where have you seen products of odd numbers divided by $k!$ before?

In the binomial expansion of $(1+y)^{-1/2}$. The coefficient of y^k is

$$\binom{-\frac{1}{2}}{k} = \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2k-1}{2}\right)}{k!} = \frac{(-1)^k}{2^k}\cdot\frac{1\times 3\times\cdots\times(2k-1)}{k!} = \left(-\tfrac{1}{2}\right)^k c_k.$$

So, taking $y = -2x$, the signs and the powers of 2 cancel:

$$(1-2x)^{-1/2} = \sum_{k=0}^{\infty}\binom{-\frac{1}{2}}{k}(-2x)^k = \sum_{k=0}^{\infty} c_k x^k = 1 + x + \frac{1\times 3}{2!}x^2 + \frac{1\times 3\times 5}{3!}x^3 + \cdots,$$

valid for $|2x| < 1$.

Now look carefully at the given series. Its general term is $c_k \cdot \frac{1}{3^{k-1}} = 3c_k\left(\frac{1}{3}\right)^k$, and it starts at $k = 2$ (the first term is $\frac{1\times 3}{2!}\times\frac{1}{3}$). So it is 3 times the series above at $x = \frac{1}{3}$, with the $k = 0$ and $k = 1$ terms removed:

$$\begin{aligned}\sum_{k=2}^{\infty} c_k \frac{1}{3^{k-1}} &= 3\left(\sum_{k=0}^{\infty} c_k\left(\tfrac{1}{3}\right)^k - c_0 - c_1\cdot\tfrac{1}{3}\right)\\ &= 3\left(\left(1-\tfrac{2}{3}\right)^{-1/2} - 1 - \tfrac{1}{3}\right)\\ &= 3\left(\sqrt{3} - \tfrac{4}{3}\right)\\ &= 3\sqrt{3} - 4.\end{aligned}$$

Here $\left(\frac{1}{3}\right)^{-1/2} = \sqrt{3}$, and $x = \frac{1}{3}$ satisfies $|2x| < 1$.

Checks

- Part (i): the partial sums 1, 2, 2.75, 3.25, 3.5625, … creep up towards 4, and summing 200 terms gives 4 to machine accuracy.
- Part (ii): $2\ln 2 \approx 1.3863$, and the partial sums 1, 1.25, 1.3333, 1.3646, … agree.
- Part (iii): the terms are 0.5, 0.2778, 0.1620, …, and summing 400 terms gives 1.196152…, matching $3\sqrt{3} - 4 \approx 1.1962$. A quick sanity check: the answer must exceed the first term 0.5 and be positive, which rules out forgetting to remove the $k = 1$ term (that would give $3\sqrt{3} - 3 \approx 2.196$, too big since the terms shrink roughly geometrically).

Common slips

- In (ii), forgetting to divide by x: $-\ln(1-x)$ at $x = \frac{1}{2}$ gives $\ln 2$, which is half the answer.
- In (iii), not noticing that the series starts at $k = 2$, or missing the extra factor of 3 from $\frac{1}{3^{k-1}} = 3 \cdot \frac{1}{3^k}$.

Beyond STEP

Differentiating and integrating a power series term by term is justified inside its radius of convergence, because a power series converges uniformly on every closed interval strictly inside that radius. This is a standard result in first-year analysis, and it is why "questions of convergence need not be considered" is safe here. The binomial series for non-integer powers is a power series of exactly this kind. Abbott's *Understanding Analysis* covers uniform convergence and power series clearly.

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